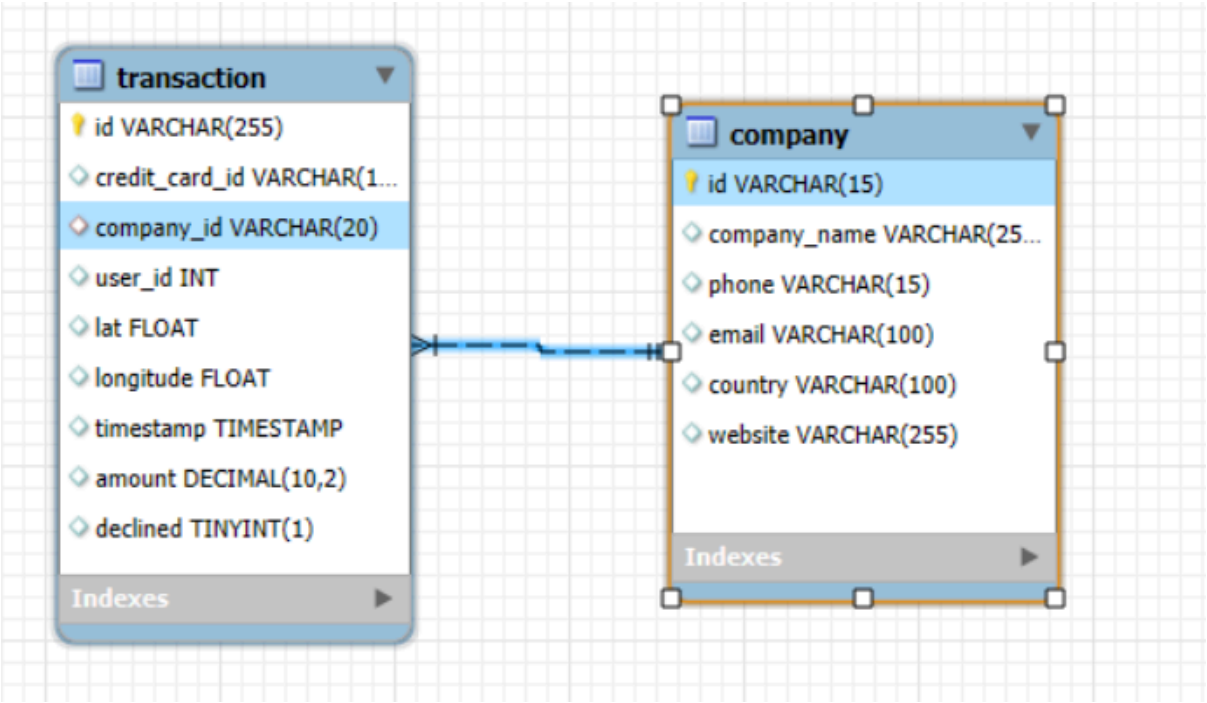
Nivell 1

Exercici 1 - A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.



Base de dades relacional composta per dues taules: company i transaction. Ambdues estan relacionades mitjançant una relació d'un (1) a molts (n). Això significa que una empresa pot realitzar múltiples transaccions, mentre que cada transacció pertany únicament a una empresa.

La relació és a través de la **foreign key** company_id de la taula transaction, que referencia a la **primary key** id de la taula company.

Exercici 2 - Utilitzant JOIN realitzaràs les següents consultes:

- Llistat dels països que estan generant vendes.

```
#EXERCICI 2 - JOINS
-- Utilitzant JOIN realitzaràs les següents consultes:
-- Llistat dels països que estan generant vendes.
SELECT DISTINCT company.country
FROM company
JOIN transaction ON company.id = transaction.company_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country
Germany
Australia
United States
New Zealand
Norway
United Kingdom
Italy
Belgium
Sweden
Ireland
China

Result 11

Output

Action Output

#	Time	Action	Message
1	11:38:29	SELECT DISTINCT company.country FROM company JOIN transaction ON company.id = transaction.company...	15 row(s) returned

- Des de quants països es generen les vendes.

```
SELECT COUNT(DISTINCT company.country)
FROM company
JOIN transaction ON company.id = transaction.company_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(DISTINCT company.country)
15

Result 15

Output

Action Output

#	Time	Action	Message
1	11:48:25	SELECT COUNT(DISTINCT company.country) FROM company JOIN transaction ON company.id = transaction....	1 row(s) returned

- Identifica la companyia amb la mitjana més gran de vendes.

```
SELECT company.company_name, AVG(transaction.amount) as vendes
FROM transaction
JOIN company ON company.id = transaction.company_id
GROUP BY transaction.company_id, company.company_name
ORDER BY vendes DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

company_name	vendes
Ac Fermentum Incorporated	284.867160

Result 16

Output

Action Output

#	Time	Action	Message
1	11:49:59	SELECT company.company_name, AVG(transaction.amount) as vendes FROM transaction JOIN company ON ...	1 row(s) returned

Exercici 3 - Utilitzant només subconsultes (sense utilitzar JOIN):

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

```
SELECT *
FROM transaction
WHERE transaction.company_id IN (select company.id
                                 from company
                                 where company.country = "Germany");
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
00138D3B-206D-4C03-94B7-63A2676EB9B4	CcS-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0
0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9	CcS-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0
00201A11-2E62-44C4-941D-198FC8DB77F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0
00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0
00687139-48B2-4FFA-8E73-B20376F04AB4	CcS-4870	b-2222	289	51.1966	10.4669	2019-03-09 19:37:49	524.84	0
0074F4DD-32F1-4827-8758-55896314623A	CcS-8081	b-2222	3500	39.7016	-8.50325	2016-12-26 23:06:57	491.90	0
00AAB9CD-39D6-4DCB-8A1D-13BE73DC90A9	CcS-6797	b-2222	2216	55.7652	-3.76245	2021-04-25 03:06:59	167.15	0
00BE09D4-6920-47D8-ABE8-325E2269829D	CcS-4983	b-2222	402	38.708	-9.12993	2019-02-27 15:25:16	141.66	0
00DA0383-E048-4577-8ED1-3C56C258FF2F	CcS-9223	b-2222	4642	51.1742	10.2027	2019-03-21 11:47:34	325.62	0
00DD11DE-ED01-4BBD-93A0-174D183A59DF	CcS-7681	b-2222	3100	45.7565	4.83109	2024-01-28 18:20:49	242.53	0
01449CE0-98E9-4DE5-9810-728C6BA00E6F	CcS-5424	b-2222	843	47.0163	2.26064	2024-02-17 19:37:14	451.71	0
0175E8C7-241E-42DA-A8B9-9F246DBF4D2F	CcS-7510	b-2222	2929	52.0619	4.29464	2021-08-28 16:29:38	9.46	0
01ABDAB8-06E2-4CA0-A131-AEE6FF11B749	CcS-5053	b-2222	472	51.7738	5.17479	2020-01-28 01:15:07	368.41	0
01F1C7ED-0823-442D-AE0E-3134D5004866	CcS-6776	b-2222	2195	59.6697	18.6697	2022-12-17 09:40:14	168.79	0
01FABC61-18C8-441B-987C-459AA3F06097	CcS-5531	b-2222	950	55.3405	-3.3863	2020-04-30 08:57:27	333.45	0

transaction 17

Output

Action Output

#	Time	Action	Message
1	11:50:59	SELECT * FROM transaction WHERE transaction.company_id IN (select company.id from company ...	2000 row(s) returned

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```
35 • SELECT company.company_name
36   FROM company
37   WHERE EXISTS (SELECT 1
38                 FROM transaction
39                 WHERE transaction.company_id = company.id
40                 AND transaction.amount > (SELECT AVG(transaction.amount)
41                                           FROM transaction));
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name
Ac Fermentum Incorporated
Magna A Neque Industries
Fusce Corp.
Convallis In Incorporated
Ante Iaculis Nec Foundation
Donec Ltd
Sed Nunc Ltd
Amet Nulla Donec Corporation
Nascetur Ridiculus Mus Inc.
Vestibulum Lorem PC
Gravida Sagittis LLP
Mus Aenean Eget Foundation

company 18

Output

Action Output

#	Time	Action	Message
1	11:53:24	SELECT company.company_name FROM company WHERE EXISTS (SELECT 1 FROM transaction ...	100 row(s) returned

- Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```
44 • SELECT *
45   FROM company
46   WHERE NOT EXISTS (SELECT *
47                     FROM transaction
48                     WHERE transaction.company_id = company.id);
49
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	company_name	phone	email	country	website
NULL	NULL	NULL	NULL	NULL	NULL

company 21

Output

Action Output

#	Time	Action	Message
1	11:56:57	SELECT * FROM company WHERE NOT EXISTS (SELECT * FROM transaction WHERE tra...	0 row(s) returned

Nivell 2

Exercici 1 - Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

```
SELECT DATE(transaction.timestamp) AS dia, SUM(transaction.amount) AS total
FROM transaction
GROUP BY dia
ORDER BY total DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

dia	total
2022-12-13	14337.44
2019-11-18	13591.32
2023-02-20	13332.59
2017-12-20	13318.43
2019-03-18	12680.95

Result 23

Output

Action Output

#	Time	Action	Message
1	12:16:20	SELECT DATE(transaction.timestamp) AS dia, SUM(transaction.amount) AS total FROM transaction GROUP B...	5 row(s) returned

Exercici 2 - Quina és la mitjana de vendes per país? Presenta els resultats ordenats de major a menor mitjà.

```
SELECT AVG(amount) as mitjana_vendes, country
FROM transaction
JOIN company ON company.id = transaction.company_id
GROUP BY country
ORDER BY mitjana_vendes DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mitjana_vendes	country
265.190742	Australia
264.977877	United States
261.153042	Belgium
260.841391	Germany
260.644761	Ireland
260.468125	Spain
259.979185	France
259.586176	New Zealand
259.375337	Norway
258.436128	Netherlands
258.272740	Italy
257.437949	Sweden
257.431229	Canada
256.759607	United Kingdom
252.422260	China

Result 25

Output

Action Output

#	Time	Action	Message
1	12:21:15	SELECT AVG(amount) as mitjana_vendes, country FROM transaction JOIN company ON company.id = transac...	15 row(s) returned

Exercici 3 - En la teva empresa, es planteja un nou projecte per a llançar algunes campanyes publicitàries per a fer competència a la companyia "Non Institute". Per a això, et demanen la llista de totes les transaccions realitzades per empreses que estan situades en el mateix país que aquesta companyia.

- Mostra el llistat aplicant JOIN i subconsultes.

```
SELECT company.company_name, transaction.id, company.country
FROM transaction
JOIN company ON company.id = transaction.company_id
WHERE company.country = (SELECT company.country
                         FROM company
                         WHERE company.company_name LIKE '%Non Institute%')
AND company.company_name != 'Non Institute';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

company_name	id	country
Sed Nunc Ltd	008629B4-C9A9-406C-A3D2-71FDA47BC546	United Kingdom
Sed Nunc Ltd	00B72BA4-54A3-4B8E-B13F-2D57535AA17A	United Kingdom
Sed Nunc Ltd	01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C	United Kingdom
Sed Nunc Ltd	023FFCE8-E618-4938-BF56-C8DF80540ADD	United Kingdom
Sed Nunc Ltd	02683BEB-EF91-4564-957B-D6F1662AB7C5	United Kingdom
Sed Nunc Ltd	02C2F29E-CEF2-4C1E-A594-F476E8F279C0	United Kingdom
Sed Nunc Ltd	02F468DC-426C-47C2-8B0A-D8B25B7A81AF	United Kingdom
Sed Nunc Ltd	0306BE3B-817B-4A49-934E-0E439291A104	United Kingdom
Sed Nunc Ltd	0347BFE6-8EB5-4387-B187-0E78E8F2B8FB	United Kingdom
Sed Nunc Ltd	03AEBD0E-DC97-4BD3-9C57-6A6DB78026FD	United Kingdom
Sed Nunc Ltd	03CA36D3-88FF-4DBF-BFD4-4CC7DA4EED2B	United Kingdom
Sed Nunc Ltd	04494182-96DD-42EB-82FE-5F92C5210537	United Kingdom
Sed Nunc Ltd	045AACF6-FF85-49FB-9DE4-E6730655366A	United Kingdom
Sed Nunc Ltd	0489FDAC-86A4-4929-954E-9A2272189CAC	United Kingdom

Result 27

Output

Action Output

#	Time	Action	Message
1	12:26:09	SELECT company.company_name, transaction.id, company.country FROM transaction JOIN company ON com...	2000 row(s) returned

- Mostra el llistat aplicant solament subconsultes.

```sql
SELECT transaction.id
FROM transaction
WHERE EXISTS (SELECT 1
              FROM company
              WHERE company.id = transaction.company_id
              AND company.country = 'United Kingdom'
              AND company.company_name != 'Non Institute');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id
008629B4-C9A9-406C-A3D2-71FDA47BC546
00B72BA4-54A3-4B8E-B13F-2D57535AA17A
01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C
023FFCE8-E618-4938-BF56-C8DF80540ADD
02683BEB-EF91-4564-957B-D6F1662AB7C5
02C2F29E-CEF2-4C1E-A594-F476E8F279C0
02F468DC-426C-47C2-8B0A-D8B25B7A81AF
0306BE3B-817B-4A49-934E-0E439291A104
0347BFE6-8EB5-4387-B187-0E78E8F2B8FB
03AEBD0E-DC97-4BD3-9C57-6A6DB78026FD
03CA36D3-88FF-4DBF-BFD4-4CC7DA4EED2B
04494182-96DD-42EB-82FE-5F92C5210537
045AACF6-FF85-49FB-9DE4-E6730655366A
0489FDAC-86A4-4929-954E-9A2272189CAC

transaction 29

Output

Action Output

#	Time	Action	Message
1	12:28:21	SELECT transaction.id FROM transaction WHERE EXISTS (SELECT 1 FROM company WHERE company.i...	2000 row(s) returned

Nivell 3

Exercici 1 - Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
SELECT company.company_name, company.phone, company.country, date(transaction.timestamp) as data_transaccio, transaction.amount as quantitat
FROM transaction
JOIN company ON company.id = transaction.company_id
WHERE transaction.amount BETWEEN 350 AND 400
AND date(transaction.timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
ORDER BY quantitat DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	phone	country	data_transaccio	quantitat
Aliquam PC	01 45 73 52 16	Germany	2024-03-13	399.84
Auctor Mauris Vel LLP	08 09 28 74 14	United States	2018-07-20	399.51
At Pede Corp.	06 14 48 33 15	Italy	2015-04-29	390.69
Aliquam PC	01 45 73 52 16	Germany	2024-03-13	388.29
Orci Adipiscing Limited	03 18 00 77 81	United Kingdom	2018-07-20	373.71
Fringilla LLC	08 29 15 93 57	New Zealand	2015-04-29	367.62
Pede Cum Ltd	07 62 26 48 38	Norway	2018-07-20	356.87
Auctor Mauris Vel LLP	08 09 28 74 14	United States	2024-03-13	353.75

Result 31

Output

Action Output

#	Time	Action	Message
1	12:32:58	SELECT company.company_name, company.phone, company.country, date(transaction.timestamp) as data_tra...	8 row(s) returned

Exercici 2 - Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen més de 400 transaccions o menys.

```
117 •  SELECT count(transaction.id) as quantitat_transaccions, company.company_name, IF(count(transaction.id) > 400, 'SI', 'NO') as mes_de_400
118    FROM transaction
119    JOIN company ON company.id = transaction.company_id
120    GROUP BY company.company_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quantitat_transaccions	company_name	mes_de_400
2401	Ac Fermentum Incorporated	SI
410	Magna A Neque Industries	SI
447	Fusce Corp.	SI
1514	Convallis In Incorporated	SI
472	Ante Iaculis Nec Foundation	SI
449	Donec Ltd	SI
1541	Sed Nunc Ltd	SI
1511	Amet Nulla Donec Corporation	SI
474	Nascetur Ridiculus Mus Inc.	SI

Result 33

Output

Action Output

#	Time	Action	Message
1	12:36:11	SELECT count(transaction.id) as quantitat_transaccions, company.company_name, IF(count(transaction.id) > 4...	100 row(s) returned